1. NAME OF THE MEDICINAL PRODUCT

Letybo Powder for Solution for Injection 100 units (Clostridium Botulinum Toxin Type A)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains
Active ingredient: Clostridium botulinum toxin type A 100 units (U) *
Stabilizer: Human serum albumin 0.50 mg,
Tonic adjuster: Sodium chloride 0.90 mg
* One unit (U) of Letybo corresponds to the calculated median lethal dose (LD_{50}) when the reconstituted product is injected intraperitoneally into mice.
Due to differences in the LD_{50} assay, these units are specific to Letybo and are not interchangeable with other botulinum toxin preparations.

3. PHARMACEUTICAL FORM

A lyophilized white powder for injection in a colorless transparent vial. The product should be colorless and transparent liquid after reconstitution with saline solution.

4. CLINICAL PARTICULARS

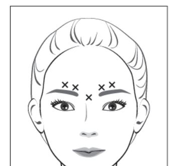
4.1 Therapeutic indications

1. Benign essential blepharospasm in adult patients aged between18 and 75
2. Temporary improvement of glabellar lines ranging from moderate to severe associated with corrugators muscle and/or procerus muscle activities in adults aged between 18 and 65
3. Muscle spasticity: Post stroke upper limb spasticity for adult patients aged 20 or above.
4. Treatment for pediatric cerebral palsy patients with dynamic equinus foot deformity by spasticity aged between 2 and 10.
5. Temporary improvement of lateral canthal lines (crow's feet lines) ranging from moderate to severe associated with orbicularis oculi muscles activity in adults aged between 19 and 65.

4.2 Posology and method of administration

1. Blepharospasm
For blepharospasm, reconstituted Letybo (see Dilution Table) is injected using a sterile, 27-30 gauge needle without electromyographic guidance. The initial recommended dose is 1.25-2.5 units (0.05 mL to 0.1 mL volume at each site) injected into the medial and lateral pre-tarsal orbicularis oculi of the upper lid and into the lateral pre-tarsal orbicularis oculi of the lower lid. In general, the initial effect of the injections is seen within three days and reaches a peak at one to two weeks post-treatment. Each treatment lasts approximately three months. At repeat treatment sessions, the dose may be increased up to two-fold if the response from the initial treatment is considered insufficient (usually defined as an effect that does not last longer than two months). However, there appears to be little benefit obtainable from injecting more than 5.0 units per site. Some tolerance may be found when this product is used in treating blepharospasm if treatments are given more frequently than every three months, and is rare to have the effect be permanent. The cumulative dose of Letybo treatment in a 30-day period should not exceed 200 units.

2. Glabellar lines



Reconstitute this product with 0.9 % preservative-free, sterile saline to make 100 units/2.5 mL (4 units/0.1 mL). Using a 30-gauge needle, inject a dose of 0.1 mL into each of 5 sites, 2 in each corrugators muscle and 1 in procerus muscle, for a total of 20 units.

In order to reduce the complication of ptosis, avoid injection near the levator palpebrae superioris, particularly in patients with larger brow depressor complexes. Injections into inner corrugators muscle and central eyebrow should be placed at least 1 cm above the bony supraorbital ridge.
Careful attention should be paid to avoid injection of this product into the blood vessel. In order to prevent exudation below the orbital ridge, be sure to firmly place the thumb or index finger below the orbital ridge, prior to injection. The needle should be toward the upper center during injection and careful attention should be paid to inject accurate volume.
Glabellar facial lines arise from the activity of corrugator muscle and orbicularis oculi muscle. These muscles move the brow medially, and the procerus muscle and depressor supercilii muscle pull the brow inferiorly. This creates a frown or "furrowed brow". The location, size, and use of the muscles vary markedly among individuals. An effective dose for facial lines is determined by gross observation of the patient's ability to activate the superficial muscles injected.
Each treatment lasts approximately three to four months. More frequent injection of this product is not recommended because the safety and efficacy are not established.

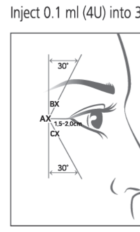
3. Muscle spasticity
Accurate dosage and number of injecting site may vary depending on the size, number, and location of relevant muscles, degree of stiffness, weakening of local muscle, and response of patient from previous treatment. Muscle stress showed clinical improvement between 3-5 weeks after injection. The amount administered in the clinical trial is shown below.

Injection site	Dosage	No. of sites
Flexed Wrist		
Flexor carpi radialis	15-60 U	1-2 sites
Flexor carpi ulnaris	10-50 U	1-2 sites
Clenched fist		
Flexor digitorum superficialis	15-50 U	1-2 sites
Flexor digitorum profundus	15-50 U	1-2 sites
Flexed elbow		
Biceps	100-200 U	Up to 4 sites

In the clinical trial, the administered amount did not exceed maximum dose 360 units, and it was injected in each muscle. A 24-30 gauge sterile syringe and a proper length needle were used for dosing, depending on the depth of the muscular tissues. The location of muscle as an injection site were selected by using electric stimulation, needle electromyography or ultrasound.

4. Dynamic equinus foot deformity by spasticity of pediatric cerebral palsy
Administer the medial and lateral heads of the rigid gastrocnemius muscle. In the case of bilateral palsy, inject to both legs at a dose of 6 units/kg body weight (3 units/kg for each leg). In the case of unilateral palsy, inject to the rigid leg at a dose of 4 units/kg body weight. The maximum dose/injection cannot exceed 200 units and after administration, the subject's status is observed for 30 minutes to detect the occurrence of acute adverse events.

5. Lateral Canthal Lines (crow's feet lines)
Lateral canthal lines arise from the activity of the orbicularis oculi muscles around the eye responsible for blinking and eyelid closure. The strong contraction of orbicularis oculi causes lateral and radially oriented folds (crow's feet lines) which originate from the lateral canthus.
The distribution of these radial lines differs among patients.
Inject 0.1 ml (4U) into 3 sites per side (6 total injection points) in the lateral orbicularis oculi muscles.



As shown in the figure, the first injection site should be at least 1.5-2.0 cm temporal to the lateral canthus and just temporal to the orbital rim (marked as AX). The next injection site (marked as BX and CX) is 1.0-1.5 cm above and below from the first injection site at an approximately 30° angle medially. The safety and efficacy of improvement of lateral canthal lines (crow's feet lines) were evaluated for 16 weeks with single dose.

6. Dilution method
Prior to injection, reconstitute this freeze-dried product with a preservative-free, sterile saline. 0.9 % Sodium Chloride Injection is the recommended diluent. Draw up the proper amount of diluent in the syringe of appropriate size. Since this product is denatured by bubbling or similar violent agitation, the diluent should be injected gently into the vial.
Discard the vial if a vacuum does not pull the diluent into the vial. Record the date and time of reconstitution on the space of the label. This product should be administered within 24 hours after reconstitution. During this period, reconstituted product should be stored in a refrigerator (2 - 8 °C). Reconstituted product should be clear, colorless and free of particulate matter. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration. Because this product and the diluent do not contain any preservative, one vial of this product should be used for a single patient.

Diluent added (0.9% Sodium Chloride Injection)	Resulting dose (U/0.1mL)
1.0mL	10.0U
2.0mL	5.0U
4.0mL	2.5U
8.0mL	1.25U

Note: These dilutions are calculated for an injection volume of 0.1 mL. A decrease or increase in dose is also possible by administering a smaller or larger injection volume from 0.05 mL (50% decrease in dose) to 0.15 mL (50% increase in dose).

4.3 Contraindications

Letybo should not be administered when;
1) The patients have known hypersensitivity to any ingredient in the formulation of Letybo.
2) The patients have neuromuscular junctional disorders (e.g. myasthenia gravis, Lambert-Eaton syndrome or amyotrophic laternal sclerosis). The diseases may be exacerbated due to the muscle relaxation activity of the drug.
3) The patients are pregnant women, women of childbearing potential, or mothers under lactation.

4.4 Special warnings and precautions for use

a) Special warnings
Since the active constituent in this drug is Clostridium botulinum toxin type A neurotoxin which is derived from *Clostridium botulinum*, the recommended dosages and frequency of administration should be observed with a full understanding of the precautions in use. Physicians administering the drug must understand the relevant neuromuscular and/or orbital anatomy of the area involved and any alterations to the anatomy due to prior surgical procedures. Understanding of standard electromyographic techniques is also required for the administration of the drug. The recommended dosage and frequency of administration for Letybo should not be exceeded.
1) Spread of Toxin Effect: The effects of botulinum toxin products may spread from the area of injection and produce negative symptoms. These may include asthenia, generalized muscle weakness, diplopia, blurred vision, ptosis, language disorder, dysphagia, dysphonia, dysarthria, urinary incontinence, and breathing difficulties. Swallowing and breathing difficulties can be life threatening and there have been reports of death. The risk of symptoms is probably greatest in children treated for spasticity but symptoms can also occur in adults treated for spasticity and other conditions, particularly in those patients who have underlying conditions that would predispose them to these symptoms. In unapproved uses, including spasticity in children and adults, and in approved indications, cases of spread of effect have occurred at doses comparable to those used to treat cervical dystonia and at lower doses.
2) Hypersensitivity reactions: Serious and/or immediate hypersensitivity reactions have been rarely reported with other botulinum toxin injections. These reactions include anaphylaxis, urticaria, soft tissue edema and dyspnea. One fetal case of anaphylaxis has been reported in which lidocaine was used as a diluent but the causal agent cannot be reliably determined. If such a reaction occurs, further injection of the drug should be discontinued and appropriate medical therapy should be immediately instituted.
3) Pre-existing neuromuscular disorders: Individuals with peripheral motor neuropathic diseases (e.g. amyotrophic laternal sclerosis, or motor neuropathy) or neuromuscular junctional disorders (e.g. myasthenia gravis or Lambert-Eaton syndrome) may be at increased risk of clinically significant systemic effects including severe dysphagia and respiratory compromise from typical doses of botulinum toxin injection. Published medical literature with other botulinum toxin injection has reported rare cases of administration of a botulinum toxin to patients with known or unrecognized neuromuscular disorders where the patients have shown extreme sensitivity to the systemic effects of typical clinical doses. In some of these cases, dysphagia has lasted several months and required placement of a gastric feeding tube.
4) Dysphagia: Dysphagia is a commonly reported adverse drug reaction following treatment of cervical dystonia patients with all botulinum toxins. In these patients, there are reports of rare cases of dysphagia severe enough to warrant the insertion of a gastric feeding tube. There are also rare case reports where subsequent to the finding of dysphagia a patient developed aspiration pneumonia and died.
5) There have also been rare reports of adverse events with other botulinum toxin injection involving the cardiovascular system, including arrhythmia and myocardial infarction, some with fatal outcomes. Some of these patients had risk factors including cardiovascular disease.
6) During the administration of other botulinum toxin product for the treatment of strabismus, retrobulbar hemorrhages sufficient to compromise retinal circulation have occurred owing to penetration of the needle into areas surrounding eyes. It is recommended that appropriate instruments to decompress the orbit be accessible. Ocular (globe) penetrations by needles have also occurred. An ophthalmoscope to diagnose this condition should be available. Inducing paralysis in one or more extraocular muscles may produce spatial disorientation, double vision or past-pointing. Covering the affected eye may alleviate these symptoms.
7) Blepharospasm: Reduced blinking from injection of botulinum toxin into orbicularis muscle can lead to corneal exposure, persistent epithelial defect and corneal ulceration, especially in patients with VII nerve disorders. In the use of other botulinum toxin product for treatment of blepharospasm, one case of corneal perforation in an aphakic eye requiring corneal grafting has occurred because of this effect. Careful testing of corneal sensation in eyes previously operated upon should be conducted and injection into the lower lid area should be avoided to reduce the risk of ectropion. Vigorous treatment of any epithelial defect should be employed. This may require protective drops, ointment, therapeutic soft contact lenses, or closure of the eye by patching or other means.
8) Lack of Interchangeability between botulinum toxin products: Since the potency units of botulinum toxin are specific to individual products, they are not interchangeable with other botulinum toxin product. Therefore, units of biological activity of botulinum toxin cannot be compared or converted into units of any other botulinum toxin product assessed with other specific assay method.
9) Injections in or near vulnerable anatomic structures: Care should be taken when injecting in or near vulnerable anatomic structures. Serious adverse events, including fatal outcomes, have been reported in patients who had received other botulinum toxin product injected directly into salivary glands, oro-lingual-pharyngeal region, esophagus and stomach. Some patients had pre-existing dysphagia or significant debility (Safety and effectiveness have not been established for indications pertaining to these injection sites). Pneumothorax associated with injection procedure has been reported following the administration of other botulinum toxin near the thorax. Caution is warranted when injecting in proximity to the lung, particularly the apices.
10) Pulmonary effects in patients with compromised respiratory status treated for upper limb spasticity or for detrusor overactivity associated with a neurologic condition: In patients with upper limb spasticity and respiratory disorder, upper respiratory tract infections and reduced lung function (decreased forced vital capacity [FVC] ≥ 15 %) have been reported more frequently compared with placebo. Reduced lung functions (decreased forced vital capacity [FVC] ≥ 15 %) have been also reported in patients with detrusor overactivity associated with a neurologic condition, having disturbances of respiratory tracts following the administration of other botulinum toxin.
11) Bronchitis and upper respiratory tract infections in patients treated for upper limb spasticity: Bronchitis was reported more frequently in patients treated with other botulinum toxin for upper limb spasticity, compared to placebo. In patients with reduced lung function treated for upper limb spasticity, upper respiratory tract infections were also reported more frequently in patients treated with botulinum toxins, compared to placebo.
b) Letybo should be carefully administered to the following patients:
1) Patients under treatment by other muscle relaxants (e.g., tubocurarine chloride, dantrolene sodium, etc.) [Muscle relaxation may be potentiated or risks of dysphagia may be increased].
2) Patients under treatments by drugs with muscle relaxing activity, e.g., spectinomycin HCl, aminoglycoside antibiotics (gentamicin sulfate, neomycin sulfate, etc.), polypeptide antibiotics (polymixin B sulfate, etc.), tetracycline antibiotics, lincomycin antibiotics (lincosamides), muscle relaxants (baclofen etc.), anti-cholinergic agents (scopolamine butylbromide, trihexylphenidil HCl, etc.), benzodiazepine and the similar drugs (diazepam, etizolam, etc.), benzamide drugs (thiapride HCl, sulpiride, etc.) [Muscle relaxation may be potentiated or risks of dysphagia may be increased].
c) General Precautions
1) This drug contains albumin, a derivative of human blood. When a drug product derived from human blood or plasma is administered into human body, the potential of infectious diseases by transmissible agents cannot be completely excluded. It may include any pathogenic agent that is still unknown. In order to minimize the risks of such infection by transmissible agents, particular cares are given to the albumin manufacturing process, including virus removal and/or inactivation process, in addition to careful screening of donors and appropriate testing of donation units.
2) Due to the nature of the disease being treated, the effects of this drug product on the ability to drive or to operate machines cannot be predicted.
3) Glabellar lines/Lateral Canthal Lines (crow's feet lines): Patients with facial palsy or ptosis symptoms, patients with infections, skin disorders or scars at proposed injection sites, patients who received facial plastic surgery, such as tissue augmentation, brow lift, a dermal resurfacing and patients who were deemed inappropriate because their glabellar lines/lateral canthal lines (crow's feet lines), were not flattened with fingers and so, their conditions could not be sufficiently improved by physical measures were excluded from the phase 3 study. Injection of this product should not be more frequent than every three months and minimum effective dose should be used.
3) Muscle Spasticity: Botulinum toxin product is only used for the treatment of localized stiffness studied in association with general standard therapeutic method. Botulinum toxin product is not effective for improving the motor range of joint being affected by a fixed rigidity.
4) Pediatric Cerebral Palsy: Botulinum toxin product is a drug product developed for treatment of local spasm in connection with standard treatments, but it is not intended to replace such treatment modalities. Botulinum toxin product is not likely to improve the motion at a joint affected by a permanent contracture.

4.5 Interaction with other medicinal products and other forms of interaction

1) The effect of botulinum toxin may be potentiated by aminoglycoside antibiotics or other drugs that interfere with neuromuscular transmission e.g. tubocurarine-type muscle relaxants. Concomitant use of Letybo with aminoglycosides or spectinomycin is contraindicated. Polymyxins, tetracyclines and lincomycin should be used with caution in the Letybo -treated patient.
2) The effects of administering different botulinum neurotoxin serotypes at the same time or within several months are unknown. Excessive neuromuscular weakness may be exacerbated by administration of another botulinum toxin before the effects of a previously administered botulinum toxin disappear.

4.6 Fertility, pregnancy and lactation

Pregnancy
There are no adequate and well-controlled studies of this product in pregnant women. In a dose study for pregnant rats, when the rats received daily intramuscular injection (1, 4, 8 U/kg/day) for 12 days during the period of organogenesis (days 5 to 16 of gestation), NOAEL (no observed adverse effect level) for mother and fetus was 1 U/kg/day or less in the both groups. In dose of 4 U/kg/day or higher, the feed intake was decreased and weight increase was inhibited. Also, the absolute weights of thymus, spleen, kidney, heart, lung and liver were decreased. Maternal toxicity is associated with increased number of dwarfism, reduced fetal body weight and fetal skeletal ossifications. However, as it does not lead to any alterations in fetal death rate, these consequences are considered as secondary effects rather than direct effects. Therefore, the potential risks observed from the rats should be apprised when administrated to pregnant women.
Lactation
It is not known whether this drug is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when this product is administered to nursing women.
Administration of this product is not recommended during pregnancy or lactation.

4.7 Effects on ability to drive and use machines

Due to the nature of the disease being treated, the effects of the drug on the ability to drive or to operate machines cannot be predicted.

4.8 Undesirable effects

a) Adverse Event
1) General
There have been rare spontaneous reports of death, sometimes associated with dysphagia, pneumonia, and/or other significant debility or anaphylaxis, after treatment with botulinum toxin. There have also been rare reports of adverse events involving the cardiovascular system, including arrhythmia and myocardial infarction, some with fatal outcomes. The exact relationship of these events to the botulinum toxin injection has not been established. The following events have been reported with other botulinum toxin injection and a causal relationship to the botulinum toxin injected is unknown: skin rash (including erythema multiforme, urticaria and psoriasiform eruption), pruritus, and allergic reaction. In general, adverse events occur within the first week following injection of the drug and while generally transient may have duration of several months. Local pain, tenderness on pressure and/or bruising, traction, swelling, hot feeling or hypertonia at injection site or adjacent muscles may be associated with the injection. Local weakness of the injected muscle(s) represents the expected pharmacological action of botulinum toxin. However, weakness of adjacent muscles may also occur due to spread of toxin. When injected in patients with blepharospasm or cervical dystonia, some distant muscles from injection site can show increased electrophysiologic jitter (rapid variation in a waveform) which is not associated with clinical weakness or other types of electrophysiologic abnormalities.
※ Incident rate is defined as follows Very rare (<0.01%), Rare (≥0.01% and <0.1%), Uncommon (≥0.1% and <1%), Common (≥ 1% and <10%), Very Common (≥10%)
2) Blepharospasm
The safety of Letybo was evaluated in 225 adult patients aged 18 or above with blepharospasm.
(Evaluation information: Double blind, randomized, active drug comparative, parallel design, phase III clinical trial, number of subjects: Letybo treatment group 121, control group 104).
Adverse events without causal relationship were observed in 38.84% (47/121 subjects, 73 cases) of Letybo treatment subjects, and 40.38% (42/104 subjects, 77 cases) of subjects in control group. The adverse drug reactions evaluated that the causal relationship with the product cannot be ruled out were observed in 28.10% (34/121 subjects, 46 cases) of Letybo treatment subjects, and 26.92%(28/104 subjects, 43 cases) of subjects in the control group. Adverse events per occurrence site are as follows and most of these adverse events were classified as mild to moderate in severity.

Letybo®

Powder for Solution for Injection 100 units (Clostridium Botulinum Toxin Type A)

[SM54000]

540

Letybo® Letybo®

Body site	Adverse events	Letybo treatment group (121 subjects) Incident rates (cases)	Control group (104 subjects) Incident rates (cases)
Eyes	Ptosis	6.61% (8 cases)	2.88% (3 cases)
	Lagophthalmos	6.61% (8 cases)	5.77% (6 cases)
	Dry eyes	6.61% (8 cases)	9.62% (10 cases)
	Watery eyes	3.31% (4 cases)	4.81% (5 cases)
	Photophobia	2.48% (3 cases)	2.88% (3 cases)
	Palpebral edema	0.83% (1 case)	0.96% (1 case)
	Chalasis	0.83% (1 case)	–
	Foreign body sensation	0.83% (1 case)	0.96% (1 case)
	Diplopia	–	0.96% (1 case)
	Entropion	–	0.96% (1 case)
	Sleep	–	0.96% (1 case)
	Keratitis	–	1.92% (2 cases)
Lymphatic system	Edema	4.96% (6 cases)	2.88% (3 cases)
Skin	Reaction at injection sites	2.48% (3 cases)	2.88% (3 cases)
	Flushing	–	0.96% (1 case)
	Itch	–	0.96% (1 case)
Pain	Myalgia	1.65% (2 cases)	0.96% (1 case)
Nervous system	Masked face	0.83% (1 case)	–

Letybo SM53800

Results of Post-Marketing Surveillance in Korea (Blepharospasm)
In the post-marketing surveillance conducted with 695 patients with benign essential blepharospasm for 6 years (2010.3.17.~2016.3.16.) in Korea, the incidence of adverse events was 21.2% (147/695, 215 cases). Among these, the incidence of adverse drug reaction of which a causal relationship with the product cannot be ruled out was 12.2% (85/695, 106 cases), which includes 2.6% of eye dryness (18/695, 18 cases), 1.9% of ptosis (13/695, 13 cases), 1.6% (11/695, 11 cases) each of ophthalmalgia and blepharoedema. Adverse drug reactions reported less than 1 % are as follow.
▪ Eye disorder: blurred vision, epiphora, diplopia, lagophthalmos, increased lacrimation, dysesthesia, lacrimal disorder, ocular irritation, lacrimation, red eye, eye strain, keratopathy, keratitis, corneal abrasion, conjunctivitis and reduced eyesight
▪ Central and peripheral nervous system disorder: headache, dizziness and ophthalmoplegia
▪ Dermal and adnexal disorder: palpebral pruritus, skin erosion, palpebral skin disease and wrinkle
▪ Platelet, hemorrhage and coagulation disorder: bruise
▪ Urinary disorder: facial edema
▪ Others: ectropion
The incidence of serious adverse events is 0.9% (6/695, 8 cases). 0.1% (1/695, 1 case) each of spine fracture, tendon rupture, hypothyroidism, hematosepsis, metrofibroma, gastritis, insomnia and pulmonary embolism were reported. No serious adverse drug reaction of which a causal relationship with the product cannot be ruled out was observed.
The incidence of unexpected adverse events was 13.1% (91/695, 127 cases), which includes 2.6% (18/695, 18 cases) of cold, 1.9% (13/695, 13 cases) of ophthalmalgia, 0.9% (6/695, 6 cases) of blurred vision, 0.7% (5/695, 5 cases) of diplopia, 0.6% (4/695, 4 cases) of ocular dysesthesia, 0.4% (3/695, 3 cases) of blepharitis, 0.4% (3/695, 3 cases) of gastritis, 0.3% (2/695, 2 cases) each of eye strain, osteoarthritis, arthralgia, spine fracture, skin erosion, dyslipidemia, bruise, facial edema, back pain and insomnia.
Unexpected adverse events reported in 0.1 % are as follow.
▪ Eye disorder: corneal abrasion, conjunctival disease, photopsia, reduced eyesight, blepharitis, macular edema
▪ Respiratory disease: sore throat, phlegm, tonsillitis
▪ Gastrointestinal disorder: dyspepsia, gastroesophageal reflux, atrophic gastritis, gastric polyp, gingivitis, hemorrhoid
▪ Musculoskeletal disorder: reduced bone mass, dysarthrosis, leg fracture, crepitation, ligament disorder, tendon rupture
▪ Dermal and adnexal disorder: palpebral pruritus, pruritus, palpebral skin disease, urticaria, onycholysis, dermatitis and wrinkle
▪ Central and peripheral nervous system disorder: ophthalmoplegia, dystonia, paralysis, Meigs' syndrome, Bell's palsy, facial pain and spinal stenosis
▪ Metabolic and nutritional disease: diabete mellitus and dystrophic calcification
▪ Defense mechanism disorder: unindicated herpes and hematosepsis
▪ Platelet, hemorrhage and coagulation disorder: pulmonary embolism
▪ Neoplasm: neoplasm of female breast and metrofibroma
▪ Vascular disorder: conjunctival hemorrhage and atherosclerosis
▪ Auditory and vestibular disorder: tinnitus
▪ Endocrine disease: hypothyroidism
▪ Erythrocyte disorder: anemia
▪ Injection site disorder: burning sensation and injection site pain
▪ Others: joint dislocation, cataract extraction, food poisoning, ectropion and laceration
Among these, the incidence of unexpected adverse events of which a causal relationship with the product cannot be ruled out was 4.6% (32/695, 35 cases). 1.6% (11/695, 11 cases) of ocular pain, 0.6% (4/695, 4 cases) each of blurred vision, diplopia, ocular dysesthesia, 0.3% (2/695, 2 cases) of facial edema, 0.1% (1/695, 1 case) each of eye strain, corneal abrasion, reduced eye sight, palpebral pruritus, skin erosion, palpebral skin disease, wrinkle, ophthalmoplegia, ectropion and bruise were reported.
The incidence of serious unexpected adverse events was 0.9% (6/695, 8 cases), which includes 0.1% (1/695, 1 case) each of spine fracture, tendon rupture, hypothyroidism, hematosepsis, metrofibroma, gastritis, insomnia and pulmonary embolism. No serious unexpected adverse drug reaction was identified.
All the outcomes of the adverse events and spontaneous reporting from the re-examination study and the adverse event report of all registered drugs in Korea (1989 - 2016.8) were evaluated at the end point of the re-examination study. No statistically significant adverse events that have not previously been reported from all other drug products were identified.
3) Glabellar lines
The safety of Letybo for use in the temporary improvement of the appearance of moderate to severe glabellar lines was evaluated in 271 adult patients aged between 18 and 65.
(Evaluation information: randomized, double blind, active control, multi center, phase III clinical trial. Number of subject: Letybo treatment group 134, control group 137)
Adverse events without causal relationship were observed in 28.4% (38/134 subjects, 63 cases) of Letybo treatment subjects, and 32.8% (45/137 subjects, 62 cases) of subjects in control group. The adverse drug reactions evaluated to that the causal relationship with the product cannot be ruled out were observed in 9.0% (12/134 subjects, 16 cases) of Letybo treatment subjects, and 11.7% (16/137 subjects, 17 cases) of subjects in the control group. Adverse events per occurrence site are as follows and most of these adverse events were classified as mild to moderate in severity.

Body site	Adverse events	Letybo treatment group (134 subjects) Incident rates (cases)	Control group (137 subjects) Incident rates (cases)
General disorder and condition of injection site	Reaction at injection sites	3.0% (4 cases)	8.0% (11 cases)
	Injection site swelling	0.7% (1 case)	–
Eye disorder	Ptosis	4.5% (6 cases)	2.2% (3 cases)
	Keratitis	0.7% (1 case)	–
	Chalasis	0.7% (1 case)	–
	Sensory disorder of eyelid	0.7% (1 case)	–
Nervous system disorder	Headache	0.7% (1 case)	0.7% (1 case)
	Dizziness	–	0.7% (1 case)
Gastrointestinal tract disorder	Vomiting	–	0.7% (1 case)
Skin and subcutaneous tissue disorder	Contact dermatitis	0.7% (1 case)	–

Results of Post-Marketing Surveillance in Korea (Glabellar lines)
In the post-marketing surveillance conducted on 815 patients with moderate or severe glabellar line wrinkles in Korea for 4 years, 6.38% (52/815, 64 cases) of adverse events occurred. Among these, reported incidence of adverse drug reactions evaluated that the causal relationship with the product cannot be ruled out was 4.79% (39/815, 45 cases), which includes 1.23% (10/815, 10 cases) of reaction in injection site, 0.98% (8/815, 8 cases) each of bruise in injected site and eyelid sensory disorder, 0.74% of pruritus in injected site (6/815, 6 cases), 0.61% of headache (5/815, 5 cases), 0.37% (3/815, 3 cases) of swelling around eyes, 0.25% (2/815, 2 cases) of pressure in injection site, 0.12% (1/815, 1 case) each of rash in injection site, swelling in injection site and ptosis. No serious adverse events and adverse drug reactions were identified.
The incidence of unexpected adverse event was 2.94% (24/815, 26 cases), which includes 0.98% (8 /815, 8 cases) of bruise in injected site, 0.37% (3/815, 3 cases) each of swelling around eyes and cold, 0.25% (2/815, 2 cases) each of rash in injected site, pressure in injected site, allergic dermatitis, 0.12% (1/815, 1 cases) each of flushing, dermatitis, chest pain, nausea, reaction in surgery site, and vaginitis. Among these, the incidence of the unexpected adverse drug reactions of which a causal relationship with the product cannot be ruled out was 1.60% (13/815, 14 cases), which includes 0.98% (8/815, 8 cases) of bruise in injected site, 0.37% (3/815, 3 cases) of swelling around eye, 0.25% (2/815, 2 cases) of pressure in injected site, 0.12% (1/815, 1 case) of rash in injected site. No serious unexpected adverse events and adverse drug reactions were identified.
All the outcomes of the adverse events and spontaneous reporting from the re-examination study and the adverse event report of all registered drugs in Korea (1989-2017.2) were evaluated at the end point of the re-examination study. No statistically significant adverse events that have not previously been reported from all other drug products were identified.
4) Upper Limb Spasticity
The safety of Letybo was evaluated in 187 adult patients aged 20 or above with post stroke upper limb spasticity.
(Evaluation information: randomized, double blind, active control, multi center, phase III clinical trial. Number of subject: Letybo treatment group 95, control group 92)
Adverse events without causal relationship were observed in 20.00% (19/95 subjects, 23 cases) of Letybo treatment subjects, and 26.09% (24/92 subjects, 30 cases) of subjects in control group. The adverse drug reactions evaluated that the causal relationship with the product cannot be ruled out were observed in 2.11% (2/95 subjects, 2 cases) of Letybo treatment subjects, and 1.09%(1/92 subjects, 1 case) of subjects in the control group. Adverse events per occurrence site are as follows and most of these adverse events were classified as mild in severity.

Body Site	Adverse events	Letybo treatment group (95 subjects) Incident rates (cases)	Control group (92 subjects) Incident rates (cases)
Infection	Rhinopharyngitis	1.05% (1 case)	–
Renal and urological disorder	Urinary frequency	1.05% (1 case)	–
Skin and subcutaneous tissue disorder	Hyperhidrosis	–	1.09% (1 case)

Results of Post-Marketing Surveillance in Korea (Upper Limb Spasticity)
In the post-marketing surveillance study of 681 patients in relation to the treatment of upper limb spasticity related to stroke in adults over 20 years of age for 4 years for re-examination in Korea, the results showed that the incidence of adverse events is 12.63% (86/681 subjects, 123 cases), regardless of causal relationship. There were no reports of serious adverse drug reactions, and serious adverse events are listed in the table below according to their frequency of occurrence.

		Serious adverse events 1.32% (9/681 subjects, 12 cases)
Occasional (≧0.1 and < 5%)	Nervous system disorder	Cerebral hemorrhage* Cerebral infarction*
	Vascular disorder	Iliac artery occlusion* Peripheral arterial occlusive disease*
	Heart disorder	Angina pectoris*
	Eye disorder	Cataract*
	Gastrointestinal disorder	Constipation*
	Infection and infestation	Virus infection*
	Injury, poisoning, and procedural complication	Upper extremity fracture*
	Mental disorder	Suicide*
	Reproductive System and Breast Disorder	Prostatitis*
	Respiratory, thoracic, and mediastinal disorder	Pneumothorax*

* Unexpected serious adverse events

In addition, unexpected adverse events and unexpected adverse drug reactions are listed in the following table according to their frequency of occurrence.

		Unexpected adverse events 10.43% (71/681 subjects, 97 cases)	Unexpected adverse events of which a causal relationship with the product cannot be ruled out 0.44% (3/681 subjects, 3 cases)
Occasional (≧0.1 and < 5%)	Musculoskeletal and connective tissue disorder	Myalgia, musculoskeletal pain, costochondritis, groin pain, knee inflammation, musculoskeletal stiffness, myofascial pain syndrome, osteopenia, pain in extremity	Myalgia
	Nervous system disorder	Headache, Dizziness, Cerebral hemorrhage, Cerebral infarction, Decreased sensation, seizure	-
	General disorder and condition of injection site	Pyrexia, Asthenia, Heat sensation, Gait disturbance, Peripheral swelling	Asthenia, Heat sensation
	Gastrointestinal disorder	Diarrhea, Abdominal pain, Lower abdominal pain, Constipation, Tooth decay, Gingival bleeding, Hemorrhoids, Nausea, Vomiting	-
	Infection and infestation	Cellulitis, Urinary tract infection, Viral infection, Adenoviral conjunctivitis, Atheroma, Onychomycosis, Periodontitis, Tuberculosis	-
	Respiratory, thoracic, and mediastinal disorder	Cough, Oropharyngeal pain, Pneumothorax, Tonsil stones	-
	Skin and subcutaneous tissue disorder	Dermatitis, Dermal cyst, Psoriasis, Seborrheic dermatitis	-
	Injury, poisoning, and procedural complication	Epicondylitis, Muscle sprain, Thermal burn, Upper extremity fracture	-
	Eye disorder	Cataract, Dry eyes, Pruritus, Ptosis	-
	Clinical examination	Abnormal liver function test, Alanine aminotransferase increase, Aspartate aminotransferase increase	-
	Reproductive System and Breast Disorder	Benign prostatic hyperplasia, prostatitis	-
	Vascular disorder	Hypotension, Iliac artery occlusion, peripheral arterial occlusive disease	-
	Heart disorder	Angina pectoris	-
	hepatobiliary disorder	Bile duct obstruction	-
	Mental disorder	Suicide	-
	Renal and urinary disorder	Urine retention	-

5) Pediatric Cerebral Palsy
The safety of Letybo was evaluated in 143 pediatric patients aged between 2 and 10 with dynamic equinus foot deformity by spasticity.
(Evaluation information: randomized, double blind, active control, multi center, phase III clinical trial. Number of subject: Letybo treatment group 73, control group 70)
Adverse events without causal relationship were observed in 73.97% (54/73 subjects, 152 cases) of Letybo treatment subjects, and 64.29% (45/70 subjects, 130 cases) of subjects in control group. The adverse drug reactions evaluated that the causal relationship with the product cannot be ruled out were observed in 2.86% (2/70 subjects, 2 cases) of Letybo treatment subjects, and they are dysphonia 1.43% (1 case) and urticaria (1 case) and all are classified as mild to moderate in severity.

Results of Post-Marketing Surveillance in Korea (Pediatric Celebral Palsy)
In the post-marketing surveillance study of 431 patients in relation to the treatment of spasticity-induced dynamic equinus foot deformity in pediatric cerebral palsy patients aged 2 years or older for 4 years in Korea, the results showed that the incidence of adverse events is 16.01% (69/431 subjects, 89 cases), regardless of causal relationship. There were no reports of serious adverse drug reactions, and serious adverse events are listed in the table below according to their frequency of occurrence.

		Serious Adverse events 1.86% (8/431 subjects, 9 cases)
Occasional (≧0.1 and < 5%)	Infection and infestation	Appendicitis* Nasopharyngitis Staphylococcal infection*
Occasional (≧0.1 and < 5%)	Nervous system disorder	Seizure* Epilepsy*
	Gastrointestinal disorder	Enteritis*

* Unexpected serious adverse events

In addition, unexpected adverse events and unexpected adverse drug reactions are listed in the following table according to their frequency of occurrence.

		Unexpected adverse events 8.12% (35/431 subjects, 44 cases)	Unexpected adverse events of which a causal relationship with the product cannot be ruled out 0.23% (1/431 subjects, 1 case)
Occasional (≧0.1 and < 5%)	Respiratory, thoracic, and mediastinal disorder	Respiratory symptom Rhinorrhea Cough Epistaxis	-
	Musculoskeletal and connective tissue disorder	Pain in extremity Back pain Muscle stress Neck mass Pain in jaw	Muscle stress
	Infection and infestation	Appendicitis Hordeolum Sinusitis Staphylococcal infection	-
	Nervous system disorder	Seizure Epilepsy Muscle rigidity	-
	General disorder and condition of injection site	Pyrexia	-
	Skin and subcutaneous tissue disorder	Acne Erythema Papule	-
	Gastrointestinal disorder	Constipation Enteritis	-
	Injury, poisoning, and procedural complication	Ligament sprain Nail injury	-
	Endocrine disorder	Precocious puberty	-
	Metabolism and nutrition disorder	Decreased appetite	-
	Renal and urinary disorder	Hypertonic bladder	-
	Vascular disorder	Extremity necrosis	-

6) Lateral Canthal Lines (crow's feet lines)
The safety of Letybo for use in the temporary improvement of the appearance of moderate to severe lateral canthal lines was evaluated in 240 adult patients aged between 19 and 65.
(Evaluation information: randomized, double blind, active control, multi center, phase I/III clinical trial. Number of subject: Letybo treatment group 119, control group 121)
Adverse events without causal relationship were observed in 13.45% (16/119 subjects, 27 cases) of Letybo treatment subjects, and 18.18% (22/121 subjects, 25 cases) of subjects in control group. The adverse drug reactions evaluated that the causal relationship with the product cannot be ruled out were observed in 1.68% (2/119 subjects, 2 cases) of Letybo treatment subjects and they are pain at the injection site 0.84% (1 case), and discomfort at the injection site 0.84% (1 case) and all are classified as mild in severity.
b) Information for patients
Letybo may cause serious side effects that can be life threatening. Call your doctor or get immediate medical help if you experience any symptoms of adverse event after the treatment with Letybo. Immediate medical help needs to be made if problems of swallowing, speaking, breathing, or muscle weakness occur. Adverse event could happen hours to weeks after the injection of Letybo. Patients with blepharospasm may have been extremely sedentary for a long time. Such patients should be cautioned to resume activity slowly and carefully after the administration of Letybo. Letybo blocks neuromuscular transmission binding to acceptor sites on motor nerve terminals, entering the nerve terminals, and inhibiting the release of acetylcholine. When injected intramuscularly at therapeutic doses, Letybo produces partial chemical denervation of the muscle resulting in a localized muscle activity reduction. In addition, the muscle may atrophy, axonal sprouting may occur, and extra junctional acetylcholine receptors may develop. There is evidence that reinnervation of the muscle may occur, thus slowly reversing muscle denervation produced by Letybo. The paralysis activity of botulinum toxin is effective for the relief of excessive abnormal contraction associated with blepharospasm. However, presence of antibodies to botulinum toxin type A may reduce the effectiveness of botulinum toxin therapy. For this reason, the dose of Letybo for blepharospasm in any case should not exceed 200 units in one-month period.
c) Pediatric use
Safety and effectiveness of this product in children and adolescent below the age of 18 years for blepharospasm and glabellar lines, below the age of 19 years for lateral canthal lines (crow's feet lines), and below the age of 20 years for muscle spasticity (post stroke upper limb spasticity) have not been established. Safety and effectiveness of dynamic equinus foot deformity by spasticity of children with pediatric cerebral palsy in children below the age of 2 years were not investigated.
d) Carcinogenesis, mutagenesis, impairment of fertility
No clinical studies have been performed to evaluate Carcinogenesis, mutagenesis and impairment of fertility.

4.9 Overdose

Signs and symptoms of overdose are not apparent immediately after injection. Should accidental injection or oral intake occur, the person should be medically supervised for up to several weeks for signs or symptoms of systemic weakness or muscle paralysis. An antitoxin may be used in the event of immediate knowledge of overdose or wrong administration. The antitoxin will not reverse any botulinum toxin-induced muscle weakness effects already appeared by the time of antitoxin administration. If the muscles of the oropharynx and esophagus are affected, aspiration may occur which may lead to development of aspiration pneumonia. If the respiratory muscles become paralyzed or sufficiently weakened, intubation and assisted respiration may be required until recovery takes place. Supportive care could involve the need for a tracheostomy and/or prolonged mechanical ventilation, in addition to other general supportive care. These patients should be considered for further medical evaluation and appropriate medical therapy immediately instituted, which may include hospitalization.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Muscle relaxants, other muscle relaxants, peripherally acting agents. ATC code: M03AX01
Mechanism of Action
Clostridium botulinum neurotoxin type A blocks the peripheral release of the neurotransmitter acetylcholine at presynaptic cholinergic nerve terminals of neuromuscular junctions by cleaving SNAP-25, a protein integral to the successful docking and release of acetylcholine from vesicles situated within the nerve endings, thereby leading to denervation of the muscle and a flaccid paralysis.
After injection, there is an initial rapid high-affinity binding of toxin to specific cell surface receptors. This is followed by transfer of the toxin across the plasma membrane by receptor-mediated endocytosis. Finally, the toxin is released into the cytosol with progressive inhibition of acetylcholine release. Clinical signs manifest within 2-3 days, with peak effect seen within 4 weeks of injection. Recovery occurs usually within 3-4 months after injection when nerve terminals sprout again and reconnect with the endplate.
Clinical efficacy
1) Phase III Clinical Trial on essential blepharospasm
Double-blinded, randomized, active control comparative, parallel-designed Phase III clinical trials (HG-06-01) were conducted to evaluate the safety and efficacy of "Letybo Inj." in essential blepharospasm.
The primary endpoint is "efficacy rate after administration of test product" and the efficacy rate is defined as 1 grade or more (Scott method) being elevated at the fourth week after administration.
When evaluated on the basis of the above criteria, the efficacy rate was 98.91% (91/92) for the test group and 100.00% for the control group. In the non-inferiority test of the test group against the control group (comparator: Botox Inj.), the lower limit of 95% confidence interval for [test group (98.91%) - control group (100.00%)] was -2.87%, which exceeded the non-inferiority boundary of -10% (d=0.10). From the Per Protocol (PP) analysis, it was confirmed that the efficacy rate of the test group was not inferior to that of the control group.

	Letybo (Test)	Botox (Comparator)
Number	91/92	82/82
Percentage	98.91%	100.00%
95% Confidence Interval (Lower limit, Upper Limit)	(97.13 , 100.00)	(100.00 , 100.00)
95 % CI (Difference between treatment groups)	-1.09 (-2.87 , 0.01)	

2) Phase III Clinical Trial on Glabellar line
Double-blinded, randomized, active control comparative, Multicenter-designed, Phase III clinical trial (HG-11-01) were conducted to evaluate the safety and efficacy for improvement of Glabellar Lines of "Letybo Inj" compared to Botox Inj. in patients with moderate to severe Glabellar Lines.
The primary efficacy endpoint was responder rate at Week 4 by physician's rating line severity at maximum frown.
Analysis with PPS (Per-Protocol Set), the main analysis population, showed that the responder rate observed in Letybo inj. group (89.34%) was greater than that observed in Botox inj. group (81.89%) by 7.45%. The lower end of two-sided 95% confidence interval for the between-group difference in the responder rate was -1.24%, which was greater than the non-inferiority margin (-14.57%); therefore, it was proved that Letybo inj. was not inferior to Botox inj.

	Letybo inj. (Test)	Botox inj. (Comparator)
Number	109/122	104/127
Percentage	89.34%	81.89%
95% Confidence Interval (Lower limit, Upper Limit)	(83.87, 94.82)	(75.19, 88.59)
95 % CI (Difference between treatment groups)	7.45 (-1.24, 16.15)	

3) Phase III clinical trial on Pediatric Cerebral Palsy
Double-blinded, randomized, active control comparative, multicenter-designed, phase III clinical trial (HG-11-02) was conducted to evaluate the safety and efficacy of "Letybo Inj." versus "Botox Inj. " in children with cerebral palsy.
The primary efficacy endpoint was the responder rate at 12-weeks after administration of the investigational product; subjects with at least a 2 point increase in PRS (Physician's Rating Scale) at Week 12 from baseline were determined as responders, and in case of bilateral palsy, subjects with at least a 2 point increase of the symptom score in each leg were determined as responders.
In the FA (Full analysis) set, the responder rate (proportion of subjects determined as responders) was 60.27% (44/73 subjects) in the Letybo Inj. group and 61.43% (43/70 subjects) in the Botox Inj.® group, and as the lower limit of 95% CI for the difference in the responder rate (-1.15%) between Letybo Inj. and Botox Inj.® was -17.16%, higher than -24.00% which was the non-inferiority margin, non-inferiority of Letybo Inj. to Botox Inj.® was demonstrated.

	Letybo inj. (Test)	Botox inj. (Comparator)
Number	44/73	43/70
Percentage	60.27%	61.43%
95% Confidence Interval (Lower limit, Upper Limit)	(49.05, 71.5)	(50.03, 72.83)
95 % CI (Difference between treatment groups)	-1.15 (-17.16, 14.85)	

4) Phase III clinical trial on Post Stroke Upper Limb Spasticity
A randomized, double blind, multi-center, active drug controlled, phase III clinical trial (HG-13-01) was conducted to compare the efficacy and safety of Letybo versus Botox in treatment of post stroke upper limb spasticity.
Primary Efficacy Assessment: Change from Baseline at Week 4 for Wrist Flexor Muscle Tone as Measured on the MAS(Modified Ashworth Scale)
In the FAS (Full Analysis set), mean MAS grade of the wrist flexor assessed by the investigator at baseline was 2.30±0.48 in the test group and 2.38±0.49 in the control group, and at Week 4 after administration, it was 0.85±0.55 in the test group and 0.98±0.54 in the control group, indicating a statistically significant decrease at Week 4 from baseline in both groups . Variation of the investigator-assessed MAS grade at Week 4 from baseline was -1.45±0.61 in the test group and -1.40±0.57 in the control group, and the variation difference between treatment groups (test group-control group) was -0.06 (95% 2-sided CI [-0.23, 0.12]), indicating no statistically significant inter-group difference. As 0.12, the upper limit of 95% 2-sided CI of the variation difference was below 0.5, the clinical non-inferiority margin, it was demonstrated that the test drug, LETYBO is not inferior to the comparator, BOTOX® .

	Letybo inj. (Test)	Botox inj. (Comparator)
Number	94	92
Baseline	2.30 ±0.48	2.38 ±0.49
Week 4	0.85 ±0.55	0.98 ±0.54
Change from baseline to Week 4	-1.45±0.61	-1.40±0.57
95 % CI (Difference between treatment groups)	-0.06 (-0.23, 0.12)	

5) Phase I/III clinical trial on Crow's feet lines
A randomized, double-blind, active-controlled, multi center, phase I/III clinical trial (HG-BOTCFL-III1)was conducted to evaluate the safety and efficacy of Letybo as compared to Botox in subject with moderate to severe crow's feet lines Primary Efficacy Assessment: Ratio of subjects whose both eyes satisfy the conditions of the improvement* of crow's feet lines when smiling as widely as possible, according to the investigator's evaluation after four weeks of the investigational product administration (* Improvement of lines: The severity of crow's feet lines when smiling as widely as possible after four weeks (Visit 3) of the investigational product administration is 0 (None) or 1 (Mild)).
As a result of the FAS analysis, improvement of crow's feet lines when smiling as widely as possible according to the investigator's on-site evaluation after four weeks of administration was 69.75 % (83/119 subjects) in the test group, and 68.33 % (82/120 subjects) in the control group. The difference in improvement rate between the test and control groups was 1.41 % [95 % two-sided confidence interval (-10.31, 13.14)], and the lower limit of the 97.5 % one-sided confidence interval (-10.31 %) exceeded -18 %, proving that the test group was non-inferior, compared to the control group.

	Letybo (Test)	Botox (Comparator)
Number	83/119	82/120
Percentage	69.75%	68.33%
95% Confidence Interval (Lower limit, Upper Limit)	(61.49, 78.00)	(60.01, 76.66)
95 % CI (Difference between treatment groups)	1.41(-10.31, 13.14)	

5.2 Pharmacokinetic properties

Due to the very small amount of product that is administered, pharmacokinetic studies are very difficult to perform and, due to the well-characterised nature of the product, are not expected to be informative. Therefore, pharmacokinetic studies have not been conducted for Letybo.
Pharmacokinetic studies have been performed using other botulinum toxins: distribution studies in rats indicate slow muscular diffusion of 125I-botulinum neurotoxin A complex in the gastrocnemius muscle after injection, followed by rapid systemic metabolism and urinary excretion. The amount of radiolabeled material in the muscle declined with a half-life of approximately 10 hours.
At the injection site, the radioactivity was bound to large protein molecules, whereas in the plasma it was bound to small molecules, suggesting rapid systemic metabolism of the substrate.
Within 24 hours of dosing, 60 % of the radioactivity was excreted in the urine. The toxin is probably metabolised by proteases and the molecular components recycled through normal metabolic pathways. Classical absorption, distribution, biotransformation and elimination (ADME) studies on the active substance have not been performed due to the nature of this product.
It is believed that at therapeutic doses, low systemic distribution of botulinum toxin occurs in patients. Clinical studies using single fibre electromyographic techniques have shown increased electrophysiologic neuromuscular activity in muscles distant to the injection site, with no associated clinical signs or symptoms.

5.3 Preclinical safety data

To evaluate the single-dose toxicity of Letybo in Sprague-Dawley (SD) rats, Letybo was administered intramuscularly to male and female rats at dose levels of 6, 30, and 150 units/kg. In conclusion, a single intramuscular administration of Letybo in male and female rats induced the paralysis of muscle, adrenergic nerve blocking syndromes, suppression of body weight gain, dark-red discharge around eyes and nose, soiled perineal region, and atrophy of thymus in an animal in female 150 units/kg group. The LD50 value was calculated to be 129.5 units/kg (95 % confidence limits; not determined). In case of the acute toxicity of Letybo in Beagle dogs, no change was found in Beagle dogs following single intramuscular administration of Letybo up to 200 units/kg group, which was equivalent to 600 times of the estimated clinical dose. Therefore, the minimum lethal dose of Letybo in Beagle dogs was higher than 200 units/kg. The No Observed Adverse Effect Level (NOAEL) in rats administered Letybo once a week for 4 weeks (5 times in total) in repeated intramuscular dose was estimated at 1.5 unit/kg unaccompanied by histological changes of muscles at the administration site. No significant toxicological change appeared in the Beagle dogs receiving Letybo once a week for 4 weeks under the test conditions. The NOAEL for Beagle dogs was 30 units/kg.
Repeated intramuscular injection of Letybo to pregnant rats in the organogenetic period resulted in an increase of paralytic gait, continuous suppression of weight gain, a decrease of body weight gain, and chronic mysitis followed by muscular atrophy in dams in groups at higher than 1 unit/kg/day dose. It induced decreases of food consumption and of organ weights of thymus, spleen, kidney, heart, lung and liver in groups at higher than 4 units/kg/day dose. In fetuses, it caused dwarfism and a decrease of fetal body weight in males at higher than 1 unit/kg/day doses, and it produced decreases of fetal body and placental weight in females, and retarded ossification characterized by a retardation of meta-tarsal ossification at higher than 4 units/kg/day/doses. Thus, under this experimental condition, the NOAEL of Letybo is considered to be less than 1unit/kg/day for both dam and embryo-fetus, and it is concluded that Letybo is not teratogenic. The impacts on embryo-fetuses are considered as secondary effects resulted by maternal toxicity.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Human serum albumin
Sodium chloride

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal.

6.3 Shelf life

The shelf-life of Letybo is 36 months from the manufacturing date.
After reconstitution, an immediate use of the solution is recommended; however, stability has been demonstrated for 24 hours at 2 – 8 °C. The vial which is expired or kept more than 24 hours after reconstitution should be abandoned.

6.4 Special precautions for storage

The unopened lyophilized vial should be stored in a refrigerator (2 - 8 °C). Reconstituted product may be stored in a refrigerator (2 - 8 °C) for up to 24 hours after reconstitution.

6.5 Nature and contents of container

Powder in a vial (Ph. Eur. type I borosilicate glass) fitted with a stopper (rubber) and a cap (aluminium), Letybo is supplied in a single use vial.

6.6 Special precautions for disposal

This unopened medicinal product should be stored in cold condition (2 - 8 °C). The reconstituted one can be stored for 24 hours in cold condition (2 - 8 °C). All vials including expired ones or any materials which have been in direct contact with the product should be disposed of as medical waste. If inactivation of toxin is required (i.e. leakage), it is recommended to use a diluted hypochlorite solution (0.5 or 1 %) before disposal as medical waste.

7. MANUFACTURER

Hugel, Inc. located at:
- 23, Geodudanji-1-gil, Dongnae-myeon, Chuncheon-si, Gangwon-do, South Korea

8. DATE OF REVISION OF THE TEXT

January 2025

312 X 360mm

312 X 360mm